Miyazaki Industrial AI Copilot — Software Requirements Specification

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1. Introduction

This document specifies a multi-tenant, Japanese-first generative-AI copilot that the Miyazaki Prefecture industrial support organisation (the Operator) would run for the manufacturing companies it supports. It is the baseline Atomic Loops and the Operator agree on before design and build start, so that scope, roles and behaviour are fixed in writing rather than discovered in code.

1.1 Purpose

The document serves three readers. The Operator uses it to confirm what the system will and will not do. Atomic Loops' product, design and engineering team uses it to design, estimate and build. Whoever accepts the delivered system uses the user stories and the non-functional requirements as the acceptance checklist.

It covers user roles, the page map, functional requirements written as user stories with acceptance criteria, page examples, the AI agent and workflow specification, the data model, non-functional requirements and a phased release scope. It does not cover commercial terms, hosting contracts or the detailed visual design.

1.2 Background

Miyazaki Prefecture runs a government-funded industrial area where manufacturing companies come to get support, and the support organisation gives member companies benefits such as new connections and contacts. During Atomic Loops' visit in October 2026, the discussion with the organisation identified one opportunity: an AI copilot that the organisation operates for its member companies.

The brief, as captured in that discussion, has eight parts; a ninth, Japanese and English throughout, was added on 23 October 2026. Appendix E maps each one to the user stories that deliver it.

1. The Operator has super-admin access and creates logins (workspaces) for each new company that joins.
2. A company enters its website and product details, and the AI analyses the products and their potential.
3. The AI then asks follow-up questions and creates follow-up workflows, so the picture of the company keeps getting more detailed.
4. The company's own page is generated by AI and holds the information about the company.
5. Many chats live in one workspace: a user can create any chat, see the workflow behind it, send messages and talk to the AI by voice.
6. A knowledge base section where the company gives prompts and the AI creates documents for them, so the system works as a copilot for the business and for business planning.
7. An option to give feedback everywhere.
8. A section where the company understands the AI use cases possible in its manufacturing plant, built on Atomic Loops' audit workflow (AI at every stage of the factory, and upstream in product

discovery and product-market fit), from which the company defines and tracks its own use cases.

9. Japanese-language interface and outcomes throughout, with English as an equal second language: every element of the dashboards, the chats and the generative AI works in Japanese and in English.

1.3 Scope

In scope for the first release (v1):

- A web application for desktop and mobile browsers with an Operator admin console and one isolated workspace per member company.
- The AI agent services behind company analysis, follow-up questions, workflows, chats, document generation and use-case recommendations.
- A Japanese-first, fully bilingual product: every interface element, dashboard, email, AI reply and generated output exists in Japanese and in English (Epic K).
- The manufacturing AI use-case library and the explainer content on the Atomic Loops method, in both languages.
- Feedback capture on every AI output and every page, with an Operator-side inbox.

Out of scope for v1 (candidates for later releases, see section 11):

- Reading data from member companies' ERP, MES or machines, and deploying or running AI use cases. The platform plans use cases; it does not execute them in the plant.
- Native mobile apps. The web app is responsive.
- Matching companies to prefecture programmes, grants and partner introductions (v2).
- Public, unauthenticated pages and self-service sign-up. Access is by invitation only.
- Payments, pricing or commercial modules.

1.4 Definitions

Term	Meaning in this document
Operator	The Miyazaki Prefecture industrial support organisation that runs the platform and onboards companies.
Member company	A manufacturing company supported by the Operator that receives a workspace.
Workspace	The isolated tenant for one member company: its profile, chats, workflows, documents and use cases.
Super Admin	An Operator staff member with rights to create workspaces, invite users and see the portfolio.
Platform Admin	An Atomic Loops administrator who maintains the use-case library, workflow templates, prompts and models.
Company Admin	The first user of a workspace (typically the company's president or a

Term	Meaning in this document
	director); manages members and sharing.
Member	Any other user in a company workspace.
Coordinator	An Operator advisor the company may grant read or comment access to its workspace.
Company profile	The AI-generated, user-verified page that holds everything the platform knows about the company.
Fact	One statement in the profile with its provenance (stated by a user, sourced from a document or website, or inferred by the AI), its confidence and its review status.
Discovery workflow	A multi-step sequence the AI creates for a company (questions, uploads, analysis, review, output) to deepen one topic.
Step	One unit of a workflow with an owner, a status and an output.
Chat	A conversation thread with the AI inside a workspace, optionally linked to a workflow or a document.
Knowledge base	The company's sources (uploaded files, crawled pages, saved chats) and the documents generated from them.
Document	A file the AI drafts from a prompt and the knowledge base, which the company edits and exports.
Use case	One AI opportunity in the company's plant or business, either taken from the library or created by the company.
Horizon	H1 quick wins, H2 strategic bets, H3 disruptive solutions: the sequence the Atomic Loops road map follows.
Autonomy level	L0 manual to L5 full autonomy, the ladder that places a use case by who decides and when.
Entry point	How a use case gets its data: software only, camera based, or machine IoT.
Atomic Assessment	Atomic Loops' five-stage factory audit: KPI alignment, stakeholder interviews and Gemba, machine workflow, document review, processing and report.
Lite assessment	The self-service, AI-guided version of that audit offered inside the platform (v1.1).

1.5 References

- Atomic Loops, "Atomic Assessment: from quick wins to autonomous operations" (the audit process and road map deck: five stages, three outcomes, four steps, L0 to L5).
- Atomic Loops, "Factory Assessment Framework" (seven operational stages with the human operations, software systems and AI opportunities at each, and the engagement framework).
- Act on the Protection of Personal Information (APPI), Japan.
- WCAG 2.1 level AA and JIS X 8341-3:2016 (web accessibility).
- OWASP Application Security Verification Standard (ASVS) level 2.

1.6 Conventions

- MUST, SHOULD and MAY carry their usual meaning. "The AI" means the platform's own agent, not a third-party assistant.
- User stories are numbered by epic (US-A01 and so on), non-functional requirements as NFR-01 and so on, diagrams as D1 and so on.
- Priority uses MoSCoW. Must and Should are in v1, Could is v1.1, Won't (v1) is a later release. The Priority column in section 4 is a dropdown so the team can re-prioritise in this document.
- Stated, Sourced and Inferred are the three provenance labels used throughout. Nothing the AI inferred is shown to a user as a fact without that label.

2. Overall description

The product is a copilot that turns the Operator's first conversation with a company into a living, AI-maintained understanding of that company, and uses it to plan the company's next steps: products, markets, the business plan and AI in the plant. The Operator sees its portfolio of companies in one place, and Atomic Loops' audit method is taught and applied through the use-case explorer.

2.1 Vision and value by stakeholder

- Member companies get a copilot that already knows their business, answers in Japanese at any hour, drafts the documents they would otherwise write by hand and shows them concretely where AI fits in their plant.
- The Operator gets a consistent onboarding path for every company, a portfolio view of who needs what, and a stronger basis for the introductions and connections it makes.
- Atomic Loops gets its method applied at scale, companies that arrive at an Atomic Assessment with a verified profile and a defined use-case list, and feedback that improves the library.

2.2 Product context

In v1 the AI never acts on the Operator's or a company's systems directly. It reads what users give it (websites, documents, answers) and writes only inside the workspace. Side-effect actions, such as creating a workspace through the Admin Copilot, always need a human confirmation.

2.3 Personas

Persona	Who they are	What they want from the platform	What slows them today	Device and language

Persona	Who they are	What they want from the platform	What slows them today	Device and language
Super Admin	Programme coordinator at the Operator	Create a workspace in minutes, see every company's status, route feedback and requests	Onboarding is a series of meetings and emails; there is no single view of the member companies	Desktop; Japanese, some English
Company Admin	President or director of a small or mid-sized manufacturer	A clear picture of the company's position and next steps; documents for banks, customers and the prefecture; a concrete view of where AI applies	Little time, limited IT staff, planning in spreadsheets and on paper	Desktop and phone; Japanese only
Member	Production, quality, planning or sales manager	Answer the follow-up questions about their area, draft specific documents, define plant use cases	Knowledge sits in people and on paper; no time to write it up	Desktop, phone on the floor; Japanese
Coordinator	Operator advisor assigned to several companies	Read a company's profile and workflows before a meeting, comment, suggest next steps	Relies on notes from past visits	Desktop; Japanese
Platform Admin	Atomic Loops product and AI team	Maintain the library, templates, prompts and evaluation sets; watch quality and cost	Content changes need a deploy unless they are editable in the console	Desktop; English

2.4 Operating environment

The platform runs as a hosted web application in a cloud region in Japan (data residency is open question OQ-03 in section 12). It supports the current and previous major versions of Chrome, Edge, Safari and Firefox on desktop, and Safari on iOS and Chrome on Android on phones. Nothing is installed on member companies' computers.

2.5 Assumptions and dependencies

- The Operator creates every account; there is no public sign-up. One Company Admin per company is invited first and then invites colleagues.
- Member companies' websites are public and crawlable. Product details may also arrive as text, PDF catalogues, spreadsheets or photographs.

- Most member companies are small or mid-sized manufacturers with limited IT staff, so the interface is simple, Japanese by default and usable on a phone.
- A large-language-model provider is available with strong Japanese capability, a contract that excludes training on customer data, and a data-processing region acceptable to the Operator (OQ-04).
- Atomic Loops supplies and maintains the use-case library, the workflow templates and the question banks; the Operator reviews the Japanese wording of that content.
- Speech-to-text and text-to-speech services with Japanese support are available for the voice feature.

2.6 Constraints

- Tenant isolation is absolute: no user, model call or retrieval ever crosses from one company's workspace into another's.
- The AI performs no action with a side effect (creating, sending, deleting) without an explicit confirmation in the interface.
- All AI output carries provenance, and numbers that were neither stated nor sourced are labelled as estimates.
- The platform complies with APPI for any personal data (user accounts, names inside documents) and keeps an audit log of administrative actions.
- The interface and all AI responses work fully in Japanese; English is a second, equal language.

2.7 Release phasing

Section 11 defines three releases: v1 (the minimum the Operator can onboard companies with), v1.1 (voice, planning canvases, the lite assessment) and v2 (programmes and connections, integrations). The priorities in section 4 follow that split.

3. Information architecture

The application has two areas, the Operator's admin console and one workspace per member company, under one shell for search, notifications, language and feedback. A Super Admin or Coordinator switches between company workspaces from the top bar; a company user only ever sees their own.

3.1 Page map

Area	Route	Page	Purpose	Who sees it
Public	/login, /invite/{token}, /reset	Sign-in, invitation, password reset	Entry to the platform; the invitation page sets language and password	Everyone
Admin	/admin	Portfolio dashboard	Companies by status, onboarding funnel, activity, feedback sentiment	Super Admin, Platform Admin
Admin	/admin/companie	Companies list	Status, profile completeness,	Super Admin

Area	Route	Page	Purpose	Who sees it
	s		last activity; create, invite, deactivate	
Admin	/admin/companies/{id}	Company record	Shared profile summary, users, workflow and use-case counts, requests, sharing settings	Super Admin, assigned Coordinator
Admin	/admin/users	Users	All Operator and company users, roles, MFA status	Super Admin
Admin	/admin/library	Use-case library	Create, edit, translate and publish use cases, stages and explainer content	Platform Admin; Super Admin reviews
Admin	/admin/templates	Workflow and document templates	Question banks, step definitions, document templates	Platform Admin
Admin	/admin/feedback	Feedback inbox	Triage, tag, reply, export	Super Admin, Platform Admin
Admin	/admin/copilot	Admin Copilot	Natural-language admin actions with confirmation (v1.1)	Super Admin
Admin	/admin/audit	Audit log	Who did what, when	Super Admin, Platform Admin
Admin	/admin/settings	Settings	Organisation name, branding, default language, retention	Super Admin
Workspace	/w/{company}/home	Home	Profile completeness, open steps, recent chats, suggested next actions	All company roles
Workspace	/w/{company}/profile	Company profile	The generated, verified profile by section; products; change history	All company roles; Coordinator sees shared sections
Workspace	/w/{company}/wo	Workflows	Active, proposed and	All company

Area	Route	Page	Purpose	Who sees it
	rkflows		completed workflows; create from template or from a goal	roles
Workspace	/w/{company}/workflows/{id}	Workflow	Stepper, step detail, linked chat, dependency view	All company roles
Workspace	/w/{company}/chats	Chats	List, search, pin; new chat	All company roles
Workspace	/w/{company}/chats/{id}	Chat	Messages, citations, context panel, voice	All company roles
Workspace	/w/{company}/knowledge	Knowledge base	Sources tab, Documents tab, Generate tab	All company roles
Workspace	/w/{company}/knowledge/documents/{id}	Document	Editor with AI actions, versions, export	All company roles
Workspace	/w/{company}/planning	Business planning (v1.1)	Canvases: business model, product-market fit, go-to-market, financial outline	Company Admin, Member
Workspace	/w/{company}/ai-use-cases	AI use-case explorer	Method, factory map, library, my use cases, readiness, road map, request	All company roles
Workspace	/w/{company}/ai-use-cases/{id}	Use case	Charter, readiness check, prerequisites, feedback	All company roles
Workspace	/w/{company}/feedback	My feedback	Everything the user submitted and its status	All company roles
Workspace	/w/{company}/settings	Workspace settings	Name, members, language, sharing with the Operator, data export and deletion	Company Admin
Global	(shell)	Search, notifications, help, feedback widget	Available on every page	Everyone

3.2 Navigation model

- A left rail carries the workspace sections (Home, Profile, Workflows, Chats, Knowledge, Planning, AI use cases); the admin console has its own rail.

- The top bar holds the company switcher (admins and coordinators only), global search, notifications, the language toggle and the user menu.
- A right-hand context panel opens on chat, workflow and document pages. It shows what the AI is using (profile, sources, workflow step) and the feedback controls.
- A persistent "Ask the copilot" button on every workspace page starts a chat with the current page as context.
- The feedback widget sits in the bottom-right corner of every page.

3.3 Role and page access

Page group	Platform Admin	Super Admin	Coordinator	Company Admin	Member
Portfolio dashboard, companies, users	Full	Full	Assigned companies, read	None	None
Library, templates, prompts	Full	Review and comment	None	None	None
Feedback inbox, audit log	Full	Full	None	None	None
Company profile	Read, for support only	Shared sections only	Shared sections, comment	Full	Full
Workflows, chats, knowledge base, planning	None	None	Read and comment, if granted	Full	Full
AI use-case explorer	Content only	Content only	Read, if granted	Full	Full
Workspace settings	None	Deactivate only	None	Full	Own profile only

"Shared sections" are the profile sections the Company Admin has chosen to share with the Operator; the default is the overview and the products summary.

3.4 End-to-end workflow and key user journeys

D3 shows the whole workflow end to end, numbered in the order a company meets it. The journeys below are the main paths through it; all follow the same pattern of AI proposal, human review and a recorded outcome.

1. J2, deepening the profile: a Member answers a workflow's questions by text or voice, the AI proposes profile updates as Stated facts, the Company Admin confirms them and the completeness score rises.

2. J3, creating a document: a user opens the Generate tab, picks a template or writes a prompt, approves the outline, edits the draft with inline AI actions and exports it. The document can be saved back as a source.
3. J4, finding AI use cases: a user reads the method explainer, opens the factory map, accepts AI-matched recommendations into "My use cases", runs the readiness check, places them on the road map and, if they wish, requests an Atomic Assessment.
4. J5, operating the portfolio: a Super Admin reviews the dashboard, opens a company record, reads and answers the feedback inbox, and uses the Admin Copilot to create the next workspace.

4. Functional requirements

Each requirement is a user story with its acceptance criteria and a priority the team can change in place. The stories are grouped into ten epics, A to J, in the order a company meets them. Section 5 shows the pages they land on and Appendix E traces them back to the eight parts of the brief.

4.1 Epic A: access, tenancy and roles

ID	Priority	User story	Acceptance criteria
US-A01	Must	As a Super Admin, I want to create a company workspace with its name, industry, size band, contact person and default language, so that a new member company is onboarded in minutes.	Workspace created with a unique slug; appears in the companies list with status Invited; the action is in the audit log.
US-A02	Must	As a Super Admin, I want to invite the Company Admin by email, so that the company can start without my help.	Invitation email in the workspace language with a link valid for 7 days; resend and revoke available; status moves to Onboarding when accepted.
US-A03	Must	As an invitee, I want to accept the invitation, set my password and choose my language, so that I can enter my workspace.	Password policy of 12 characters or more enforced; language saved to the user; the onboarding wizard opens on first entry.
US-A04	Must	As a Company Admin, I want to invite colleagues as Members, so that the people who know each area can contribute.	Invite by email with role Member; remove a member; pending invitations listed with resend.
US-A05	Must	As any user, I want sign-in, sign-out, password reset and an idle timeout, so that my account stays safe.	Reset link valid for 1 hour; session ends after 12 hours idle; repeated failed sign-ins are rate-limited and logged.

ID	Priority	User story	Acceptance criteria
US-A06	Should	As an admin, I want multi-factor authentication with an authenticator app, required for admin roles, so that admin accounts are protected.	Enrolment forced at first sign-in for Super Admin and Platform Admin; recovery codes issued; MFA status shown in the users page.
US-A07	Must	As the platform, I must keep every workspace's data isolated, so that no company ever sees another's.	Every query is scoped by workspace on the server; automated tests prove cross-workspace requests are refused; retrieval indexes are partitioned per workspace.
US-A08	Should	As a Company Admin, I want to grant a named Coordinator read or comment access, so that our advisor is prepared before meetings.	Grant and revoke from workspace settings; the grant is in the audit log; the Coordinator can read and comment but not edit.
US-A09	Must	As any user, I want to switch the interface between Japanese and English, so that I work in my own language.	Interface, emails and AI replies follow the user's language; a user can ask the AI for one reply in the other language.
US-A10	Must	As a Super Admin, I want to deactivate and reactivate a workspace, so that a company that leaves the programme loses access without losing data.	Deactivated users cannot sign in; data is kept for the retention period; reactivation restores everything unchanged.

4.2 Epic B: admin console

ID	Priority	User story	Acceptance criteria
US-B01	Must	As a Super Admin, I want a companies list with status, profile completeness, use-case count and last activity, so that I see who needs attention.	Statuses Invited, Onboarding, Active, Dormant (no activity for 30 days), Deactivated; filter, sort and CSV export.
US-B02	Must	As a Super Admin, I want a company record with the shared profile summary, users, workflow and document counts and open requests, so that I can prepare a conversation.	Shows only sections the company has shared; counts never expose content; links to the feedback and requests from that company.
US-B03	Must	As a Super Admin, I want a portfolio dashboard, so that I can	Counts by status; onboarding funnel (invited, accepted, profile approved, first workflow done); weekly active

ID	Priority	User story	Acceptance criteria
		report on the programme.	companies; industries; most-chosen use cases; feedback sentiment; all aggregated and respecting sharing settings.
US-B04	Must	As a Platform Admin, I want to manage the use-case library, so that content improves without a deploy.	Create and edit use cases with Japanese and English fields, stage, entry point, horizon, autonomy level and data needed; draft, review, published states; version history.
US-B05	Should	As a Platform Admin, I want to manage workflow templates and question banks, so that discovery improves over time.	Steps, questions, branching rules and outputs editable in both languages; publish creates a new template version; running workflows keep their version.
US-B06	Must	As a Super Admin, I want a feedback inbox, so that every piece of feedback is seen and answered.	All feedback with type, page and a reference to the AI output; statuses New, In review, Addressed; tags; reply to the user; export.
US-B07	Must	As a Super Admin, I want an audit log of admin actions and of any Operator access to company data, so that access is accountable.	Immutable entries with actor, action, target and time; filter by actor, company and date; retained for 1 year.
US-B08	Could	As a Super Admin, I want an Admin Copilot that carries out admin requests in natural language, so that routine administration takes seconds.	The AI shows the exact action it will take; nothing executes before confirmation; every action is logged; portfolio questions are answered from aggregated data only.
US-B09	Could	As a Super Admin, I want to post announcements, so that every company hears about events and deadlines.	Announcement shown on every workspace home; included in the email digest; can be scheduled and withdrawn.
US-B10	Won't (v1)	As a Super Admin, I want to record programmes, events and partners and let the AI match companies to them, so that the connections the Operator makes are systematic.	Deferred to v2; companies opt in to matching; matches are suggestions the Operator confirms.

4.3 Epic C: onboarding and the company profile

ID	Priority	User story	Acceptance criteria
US-C01	Must	As a Company Admin, I want an onboarding wizard that takes our website, product details and basic facts, so that the AI can start building our profile.	Fields for website URLs, product text, catalogue and photo uploads, employees, revenue band, plants, certifications, industries served and customer types; progress saved; resumable.
US-C02	Must	As a Company Admin, I want the AI to read our website and extract what it learns, so that I do not retype what is already public.	robots.txt respected; depth and page limits applied; every extracted fact carries its source URL; live progress; completes within 5 minutes for sites up to 200 pages; a failed fetch shows a clear message and a manual entry fallback.
US-C03	Must	As a Company Admin, I want the AI to analyse each product's function, process, customers, differentiators and potential, so that we see our products the way a market analyst would.	One entry per product or product family with provenance and confidence per statement; the user can add, merge, rename or delete products; adjacent applications and potential markets are labelled Inferred.
US-C04	Must	As a Company Admin, I want to confirm, edit or reject every fact on the generated profile, so that the profile only holds what is true.	Sections: Overview, Products and services, Markets and customers, Manufacturing capabilities, Supply chain and partners, Competitive position, Potential and opportunities, Open questions; each fact shows Stated, Sourced or Inferred and a confidence; rejected facts leave the profile and the AI context; every change is logged with the editor and time.
US-C05	Must	As a Company Admin, I want a completeness score and a gap list, so that I know what to answer next.	Score per section and overall; each gap names the workflow or question that would fill it; the home page shows the top three gaps.
US-C06	Must	As a Member, I want facts I state in chats, answers and documents to flow into the profile after review, so that the profile keeps up with what we tell the AI.	Proposed facts queue with provenance; the Company Admin confirms or rejects; per-fact history with before and after values.
US-C07	Should	As a Company Admin, I want to re-run the website analysis, so that the profile follows changes	A diff of new, changed and removed facts is shown; confirmed facts are never

ID	Priority	User story	Acceptance criteria
		to our site.	overwritten without review.
US-C08	Must	As a Company Admin, I want to choose which profile sections the Operator can see, so that we control what leaves the company.	Default: overview and products summary; changes logged; shared sections carry a "visible to Operator" marker.
US-C09	Should	As a Company Admin, I want to export the profile, so that I can send it to a bank or a customer.	DOCX and PDF in Japanese or English; the export carries the as-of date and the provenance legend.
US-C10	Must	As the platform, I must treat website and document content as data, never as instructions, so that a crafted page cannot change the AI's behaviour.	Instruction-like text found in sources is ignored and flagged on the source; a red-team test set of injected pages passes before each release.

4.4 Epic D: discovery workflows

A workflow is how the AI turns a gap in the profile into a short, structured piece of work for the company. The AI proposes it, the company runs it, and its outputs land in the profile, the knowledge base or the use-case list.

ID	Priority	User story	Acceptance criteria
US-D01	Must	As a Company Admin, I want the AI to propose the first workflows once our profile exists, so that we know where to start.	Proposals: Company discovery, one Product deep-dive per main product, Market and product-market fit, Manufacturing and AI readiness; each with a rationale, the number of questions and uploads involved; the user starts, defers or dismisses each.
US-D02	Must	As a Member, I want a workflow to be a clear sequence of steps with a status and an owner, so that I always know what is next.	Step types: question set, upload request, AI analysis, review and approval, output document; statuses Not started, In progress, Waiting for review, Done, Skipped; owner and optional due date per step.
US-D03	Must	As a Member, I want the AI to ask questions in small, adaptive sets, so that answering feels like a conversation rather than a form.	5 to 10 questions per set; follow-ups depend on earlier answers; answer by text, voice or upload; "I don't know" and "assign to a colleague" available; answers become Stated facts in the review queue.

ID	Priority	User story	Acceptance criteria
US-D04	Must	As a Company Admin, I want to see a workflow as a stepper and as a dependency view, so that I understand what each step feeds.	Stepper with progress; dependency view shows which profile sections, documents and use cases each step feeds and which steps wait on which.
US-D05	Must	As a Member, I want every workflow to have its own chat, so that I can ask the AI about the step I am on.	Messages carry the workflow context; the AI can mark a step done or move to the next one from the chat after confirmation.
US-D06	Should	As a Company Admin, I want to create my own workflow from a template or by describing a goal, so that the AI works on what matters to us.	The AI drafts the steps from the goal; the user reorders, adds, removes and assigns steps before starting.
US-D07	Should	As a step owner, I want reminders and the ability to pause and resume, so that workflows do not stall silently.	Notification when a step waits on me; weekly digest of open steps; pause and resume keep all answers.
US-D08	Must	As a Company Admin, I want a completed workflow to deliver its outputs and propose what comes next, so that momentum continues.	Outputs produced as defined by the template (profile updates, a document, use-case candidates); next workflows proposed with rationale.
US-D09	Should	As a Member, I want to turn any chat into a workflow, so that a good conversation becomes structured work.	"Turn this into a workflow" drafts steps from the conversation; the user edits before starting.

4.5 Epic E: chats and voice

ID	Priority	User story	Acceptance criteria
US-E01	Must	As a Member, I want a list of chats I can search, pin, rename and archive, and to start a chat blank, from a template, from a workflow or from a document, so that I find and start conversations quickly.	List sorted by last activity; search across titles and content; a chat started from a workflow or document opens with that context attached.
US-E02	Must	As a Member, I want replies that stream in, with formatting and citations, and actions on every	Streaming starts within 2 seconds at p95; markdown and tables render; citations link to profile facts, sources or web

ID	Priority	User story	Acceptance criteria
		message, so that I can use what the AI says.	pages; actions: copy, feedback, save to knowledge base, turn into document, turn into workflow, propose to profile.
US-E03	Must	As a Member, I want to see and control what context the AI uses, so that I trust its answers.	Context panel lists the profile sections, sources and workflow in use as toggles; the AI states when it answers without the profile.
US-E04	Could	As a Member, I want to talk to the AI by voice, so that I can use it on the floor or while driving.	Push-to-talk and hands-free modes; Japanese and English recognition; optional spoken replies; transcript saved as the message text; works on a phone browser.
US-E05	Must	As a Member, I want to attach files in a chat, so that the AI can read a drawing, a price list or a report.	PDF, images, spreadsheets, Word and PowerPoint up to 25 MB per file; the attachment can be saved as a source.
US-E06	Should	As a Member, I want suggested prompts based on our situation, so that I know what the AI can do next.	Suggestions drawn from open gaps, waiting steps and newly relevant library use cases; dismissible.
US-E07	Must	As a Company Admin, I want facts we state in chat to be captured for the profile, so that nothing we say is lost.	The AI proposes recognised facts to the review queue with the message as source; nothing enters the profile without confirmation.
US-E08	Should	As a Member, I want to share a chat with colleagues and export it, so that the team can build on it.	Share with members; export as Markdown or PDF; a shared chat shows who said what.
US-E09	Must	As a Super Admin, I want a monthly token budget per workspace, so that costs stay predictable.	Default set in admin settings; the workspace sees a notice at 80% and a clear message at the limit; the Super Admin can raise it.
US-E10	Must	As a Member, I want long chats to keep working, so that a conversation can run for weeks.	Older messages are summarised into the context; the summary is visible in the context panel.

4.6 Epic F: knowledge base and document studio

ID	Priority	User story	Acceptance criteria

ID	Priority	User story	Acceptance criteria
US-F01	Must	As a Member, I want to add files and web pages as sources, so that the AI knows what we know.	PDF, DOCX, XLSX, PPTX, images, text and URLs; OCR for scanned Japanese PDFs; statuses Processing, Ready, Failed; automatic tags; 1 GB per workspace in v1, configurable.
US-F02	Must	As a Member, I want to open a source and see what the AI extracted, so that I can check and clean it.	Extracted text, AI summary and detected facts shown; re-index available; deleting a source removes it from retrieval within one minute.
US-F03	Must	As a Member, I want to generate a document from a prompt and a template, so that a first draft takes minutes.	Template or free prompt; context chosen from profile, selected sources and a workflow's answers; language and length chosen; the request is saved for re-use.
US-F04	Should	As a Member, I want to approve an outline before the full draft, so that the document has the structure I need.	Outline shown first; sections can be added or removed; each section can be regenerated on its own.
US-F05	Must	As a Member, I want an editor with AI actions on selected text, so that I finish the document without leaving it.	Rich text; rewrite, shorten, expand, translate between Japanese and English, insert profile facts, build a table from sources; comments; versions with compare and restore.
US-F06	Must	As a Member, I want to export and share a document, so that it reaches banks, customers and the Operator.	DOCX, PDF and Markdown export; share with members and, if granted, the Coordinator.
US-F07	Must	As a Member, I want to save a finished document as a source, so that later chats and documents build on it.	One action adds the document to the knowledge base and indexes it; the source shows its origin.
US-F08	Should	As a Platform Admin, I want to maintain document templates, and as a Company Admin to save our own, so that drafts match how we write.	Platform templates in both languages; a company template saved from any document; templates versioned.
US-F09	Must	As a Member, I want every statement in a generated	Sourced statements link to the source; unsourced statements carry an "AI

ID	Priority	User story	Acceptance criteria
		document to show whether it is sourced, so that I know what to verify.	assumption, verify" marker; the export can include or exclude the markers.
US-F10	Could	As a Member, I want to translate a whole document between Japanese and English, so that one draft serves both audiences.	Workspace glossary keeps product and process terms consistent; the translation becomes a new version.

4.7 Epic G: business planning copilot (v1.1)

ID	Priority	User story	Acceptance criteria
US-G01	Could	As a Company Admin, I want planning canvases pre-filled from our profile, so that planning starts from what the AI already knows.	Business model canvas, product-market fit board (segment, problem, product, alternatives, evidence, status), go-to-market plan, three-year financial outline, export readiness; each cell editable; AI fills carry provenance.
US-G02	Could	As a Company Admin, I want the AI to challenge a canvas, so that weak assumptions are found before a bank or a customer finds them.	A "challenge" action lists the weakest assumptions, why, and the evidence that would resolve each.
US-G03	Could	As a Member, I want each hypothesis to link to its evidence, so that we track what is proven.	Links to sources, chats and workflow answers; statuses Untested, Validated, Invalidated.
US-G04	Could	As a Company Admin, I want an action plan derived from the canvases, so that planning turns into tasks.	Tasks with owner and due date; progress shown on home.
US-G05	Could	As a Company Admin, I want to export the planning pack, so that it can be presented or submitted.	One document from all canvases in the chosen language, through the document studio.

4.8 Epic H: manufacturing AI use-case explorer

This is where the Atomic Loops method lives inside the product. The explorer teaches how AI fits a factory, shows use cases stage by stage, matches them to the company's profile and lets the company define its own, in the same horizon and autonomy language that an Atomic Assessment report uses. Use cases span every stage of the factory flow and extend upstream into product discovery and product-market fit; the library is not a fixed list and grows as Atomic Loops adds to it.

ID	Priority	User story	Acceptance criteria
US-H01	Must	As a Company Admin, I want a short explainer of how AI fits a factory, so that I can talk about it with my team.	Covers the five audit stages, the three outcomes of a report, the four steps (descriptive, predictive, prescriptive, autonomous), the L0 to L5 ladder, the three horizons and the value principles (measured before modelled, targets derived, haircuts applied, overlaps combined); each concept has one example; Japanese and English; readable in under 15 minutes.
US-H02	Must	As a Member, I want a factory map I can click through stage by stage, so that I see where AI applies in a plant like ours.	Seven operational stages (inward, warehouse, kitting, assembly, final inspection, packaging, logistics) plus lanes for planning and supply chain, maintenance, quality, R&D and product discovery, and sales and product-market fit; selecting a stage shows typical human operations, software systems and example use cases; stages indicated by our profile are highlighted.
US-H03	Must	As a Member, I want each library use case as a card I can search and filter, so that I can compare options.	Card fields: title, stage, problem solved, how it works, data needed, systems touched, entry point (software only, camera based, machine IoT), horizon, autonomy level, prerequisites, what good looks like; value shown only as observed ranges with no client named; Japanese and English; filters by stage, entry point, horizon and level.
US-H04	Must	As a Company Admin, I want the AI to recommend use cases that fit our profile, so that we do not read the whole library.	Ranked list with a rationale per use case tied to profile facts and the readiness gaps it would face; "add to my use cases" in one click; feedback on every recommendation.
US-H05	Must	As a Member, I want to define our own use case with the AI's help, so that ideas from the floor become something we can plan.	Wizard: problem, stage, who is affected, how the data is captured today (paper, spreadsheet, system, sensor), systems involved, expected outcome and KPI; the AI drafts a one-page charter, classifies the step and autonomy level and lists prerequisites and data needs; the user edits and saves.
US-H06	Should	As a Company Admin, I want a readiness check per use case, so that we know what has to exist before it can run.	Four dimensions (data, process, people, systems) with 3 to 5 questions each; a readiness rating and gap list; the level the use case needs against where we are today.

ID	Priority	User story	Acceptance criteria
US-H07	Must	As a Company Admin, I want a road map board, so that our use cases are sequenced the way the method says.	A 2x2 of business impact against readiness to deploy with quadrants Quick wins, Strategic bets, Momentum builders and Later; a programme view by horizon (H1 to H3) on a multi-year line; drag and drop; dependencies shown as "needs the data created by"; export as a document.
US-H08	Could	As a Company Admin, I want a self-service lite assessment, so that we can prepare the ground before a full audit.	Five stages in self-service form: KPI alignment questions, a pain-point interview in chat, stage-by-stage capture of each line or machine with data, human and material lanes, a document checklist, and a preliminary readiness summary with recommended quick wins; every output is labelled preliminary.
US-H09	Could	As a Company Admin, I want to request an Atomic Assessment from inside the platform, so that the next step is one click away.	Form pre-filled from the profile, use cases and readiness; routed to Atomic Loops and the Operator; statuses Requested, Scheduled, In progress, Report delivered visible to the company.
US-H10	Must	As a Member, I want learning content per stage, so that I understand each use case before I commit to it.	Short explainer, an anonymised example and the "what we need from the plant" list per stage; a glossary in both languages.

4.9 Epic I: feedback everywhere

ID	Priority	User story	Acceptance criteria
US-I01	Must	As any user, I want to rate any AI output in one click and say why, so that the AI improves on what matters to us.	Thumbs up or down, a reason from a short list and free text on every message, profile fact, document, recommendation and charter; non-blocking; stored with a reference to the output.
US-I02	Must	As any user, I want a feedback widget on every page, so that I can report anything, anywhere.	Categories: bug, idea, confusing, wrong, praise; free text; optional screenshot; page, role and language captured

ID	Priority	User story	Acceptance criteria
			automatically.
US-I03	Must	As a Member, I want to report a wrong fact where I see it, so that the profile is corrected quickly.	The fact is marked Disputed; the correct value is requested; the item goes to the review queue and the inbox.
US-I04	Must	As any user, I want to see what happened to my feedback, so that giving it feels worthwhile.	Statuses Received, In review, Addressed; the reply from the Operator or Atomic Loops shown under the item.
US-I05	Must	As a Super Admin, I want feedback analytics, so that I can report on satisfaction and spot problem areas.	Volume, positive rate, by feature and by company in aggregate, trend over time; export.
US-I06	Must	As a Platform Admin, I want feedback and the related output in the evaluation set, so that prompts and models are tuned on real cases.	Available only where the company's consent setting allows; never used to train third-party models; personal data removed before use.
US-I07	Should	As a Super Admin, I want a short quarterly in-app survey, so that we measure satisfaction consistently.	Three questions, once per quarter per user, dismissible; results on the dashboard.

4.10 Epic J: notifications, activity, settings and help

ID	Priority	User story	Acceptance criteria
US-J01	Must	As any user, I want notifications in the app and by email digest, so that I act when something waits on me.	Events: invitation, step waiting on me, document ready, feedback answered, announcement; per-user preferences; daily or weekly digest.
US-J02	Should	As a Company Admin, I want an activity feed, so that I see what the team and the AI changed.	Facts confirmed, documents created, use cases added, workflows started and completed, with actor and time.
US-J03	Must	As a Company Admin, I want workspace settings, so that we control our data and our team.	Name, logo, default language, members, Coordinator access, sharing with the Operator, data retention, export of all data (documents, profile, chats as JSON and Markdown in a ZIP) and a deletion request.
US-J04	Must	As a new user, I want a guided tour and help, so that I am productive on day one.	First-run tour, help articles in both languages, contextual tips, an onboarding checklist on home that ticks itself off.

ID	Priority	User story	Acceptance criteria
US-J05	Should	As any user, I want to search across everything in the workspace, so that I find what the AI and the team produced.	Profile, chats, documents, sources and use cases searchable with filters by type and date.
US-J06	Must	As the platform, I must record consent, so that personal data and feedback are used lawfully.	Terms and privacy notice accepted at first sign-in; consent to Operator visibility and to feedback use are explicit settings with their change history.

4.11 Epic K: Japanese first, English equal

Member companies work in Japanese; the Operator and Atomic Loops work in both languages. Every element of the product, from the navigation to the dashboards, the chats, the generated profile and the documents, exists in Japanese and in English, and a user never meets an untranslated screen. NFR-13 states the platform-level rule; the stories below make it testable.

ID	Priority	User story	Acceptance criteria
US-K01	Must	As any user, I want every interface element in Japanese and in English, so that I never meet an untranslated screen.	Navigation, buttons, forms, tables, charts, dashboards, emails, error messages and help exist in both languages; Japanese is the default for company workspaces; the language toggle is on every page; an automated test fails the build on any missing string; screenshots of every page in both languages are reviewed before release.
US-K02	Must	As a Member, I want the AI to reply in my language and switch when I ask, so that the chat feels native.	Replies follow the user's language; a chat can be switched to the other language; a mixed-language message (a Japanese question with English product names) is handled; a single reply in the other language is available on request.
US-K03	Must	As a Company Admin, I want every generated output in the workspace language and available in the other language on demand, so that the same material serves Japanese and overseas readers.	Profile, facts, documents, charters, recommendations and the road map are generated in the workspace language; a one-click version in the other language is produced; the workspace glossary and Appendix F keep terminology consistent across both.
US-K04	Must	As a Company Admin, I want the AI to read our Japanese	Japanese encodings, vertical text in PDFs where feasible and Japanese OCR are

ID	Priority	User story	Acceptance criteria
		website and documents correctly, so that the profile is built from what we actually publish.	handled; facts are stored in the source language and shown with a translation labelled "translated" when the viewer's language differs.
US-K05	Must	As a Platform Admin, I want the library and the explainer content to be bilingual by construction, so that nothing reaches a company in one language only.	Every use case, stage and explainer has Japanese and English fields; publishing is blocked until both exist; the Operator's reviewer approves the Japanese wording (OQ-10).
US-K06	Must	As any user, I want dashboards and charts labelled in my language with local formats, so that numbers read correctly.	Axis labels, legends, statuses and tooltips in both languages; numbers, dates and currency formatted per locale, yen formatted per locale; ISO dates in admin exports.
US-K07	Should	As a Member, I want search and retrieval to work across languages, so that an English source answers a Japanese question.	A query in either language retrieves sources in both; citations show the source's original language.
US-K08	Could	As a Member, I want voice to work in Japanese and in English without choosing first, so that I can just speak.	Language detected per utterance; transcript stored in the spoken language; spoken replies in the user's language (v1.1).
US-K09	Must	As a Japanese user, I want typing and reading to behave correctly, so that the interface feels like a Japanese product.	IME composition handled in every field, including the chat composer; Japanese line-breaking rules; fonts with full JIS coverage on web and in exports.
US-K10	Should	As a Super Admin or Coordinator, I want to read a company's shared content in my own language, so that I can prepare without a translator.	Shared sections and feedback shown in the viewer's language with a "translated by AI" marker where the original differs; exports keep the original language unless a translation is requested.
US-K11	Must	As any user, I want notifications and emails in my language, so that nothing arrives in a language I do not read.	Every template exists in both languages; the recipient's language setting decides; the invitation email follows the workspace default until the user chooses.

5. Page examples

The wireframes below fix the layout and the components of the key pages; the visual design comes later. Labels are shown in English for readability and are Japanese in the product. The sample company, Sample Co. A, is fictional: a 48-person precision machining company making hydraulic valve bodies and manifolds for construction-equipment makers. All figures in the samples are illustrative.

5.1 Admin console: companies list and create company

The Super Admin's working page: every company, its status and whether it needs attention, with workspace creation one click away (US-A01, US-B01).

```
[Operator]  Admin console        Search...   [Bell 3]  JA | EN  User

Dashboard     Companies (23)          Filter: All ▾    [+ Create]
Companies ●
Users         ┌──────────────┬──────────┬──────────┬─────────┬──────────┐
Library       │ Company      │ Status   │ Profile  │ Use     │ Last     │
Templates     │              │          │ complete │ cases   │ active   │
Feedback      ├──────────────┼──────────┼──────────┼─────────┼──────────┤
Copilot       │ Sample Co. A │ Active   │ 78%      │ 4       │ 2 h      │
Audit log     │ Sample Co. B │ Onboarding│ 35%     │ 0       │ 1 d      │
Settings      │ Sample Co. C │ Invited  │ -        │ -       │ -        │
              │ Sample Co. D │ Dormant  │ 91%      │ 7       │ 34 d     │
              └──────────────┴──────────┴──────────┴─────────┴──────────┘
              Export CSV                                  < 1 2 3 >
```

```
┌ Create company ──────────────────────────────────────────┐
Company name (JA)   [                                    ]
Company name (EN)   [                                    ]
Industry            [ Metal processing            ▾ ]
Size band           [ 21 to 50 employees          ▾ ]
Default language    (●) Japanese   ( ) English
Company Admin       Name [           ]  Email [       ]
Coordinator         [ none                        ▾ ]
                          [Cancel]   [Create and invite]
└──────────────────────────────────────────────────────────┘
```

States: empty list with a "create your first company" prompt; row actions (open, resend invitation, deactivate); a Dormant badge after 30 days without activity.

5.2 Onboarding wizard

Four steps the Company Admin completes on first entry: language, website and products, basic facts, review (US-C01 to US-C03). Step 2 is shown, followed by the analysing state.

```
Welcome, Sample Co. A        Step 2 of 4: Website and products
○──────────●──────────○──────────○
Language  Website   Facts    Review

Your website
[ https://www.example.co.jp                    ]  [+ Add another]
```

```
Your products (describe them in your own words, or upload a
catalogue)

 We machine hydraulic valve bodies and manifolds for
 construction-equipment makers, lot sizes 50 to 500 ...

[ Upload catalogue (PDF, XLSX) ]  [ Add photos ]  [ Speak instead ]

Products found so far
- Hydraulic valve bodies        The AI will read your website
- Manifolds                     and catalogue, then ask a few
- Pump housings     [+ Add]     questions. Nothing is shared
                                outside your workspace.

                                              [Back]   [Analyse ->]
```

```
Analysing your company ...
✓ Read 38 pages from www.example.co.jp
✓ Found 3 product families, 2 certifications, 1 plant
● Analysing products and potential ...
○ Preparing your profile
Found so far:   ISO 9001                 (sourced: /company/quality)
                5-axis machining centres (sourced: /equipment)
```

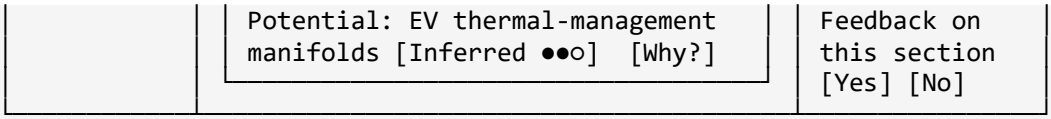
States: a site that cannot be fetched shows the reason and a "describe your company instead" path; the analysis can run in the background while the user continues to step 3; the wizard resumes where it stopped.

5.3 Company profile

The living page the brief calls the company's page (US-C04 to US-C08). Every fact carries its provenance and the controls to confirm, edit or reject it; the right-hand column turns gaps into the next workflow.

```
Sample Co. A  >  Profile                Completeness 62%   [Export ▾]

Overview ●   | Overview                Visible to        | Gaps and next
Products     |                         Operator ✓        | steps
Markets      |  48 employees, Miyazaki  [Stated]         | 1 Which three
Capabili-    |  Founded 1987            [Sourced >]      |   customers
 ties        |  Precision machining of  [Sourced >]      |   bring most
Supply       |  hydraulic components                     |   revenue?
 chain       |  Likely tier-2 supplier  [Inferred        |   -> Workflow:
Competi-     |  to two or three OEMs     ●●○ 70%]        |   Markets
 tion        |    [Confirm] [Edit] [Reject]              | 2 Capacity
Potential    |                                           |   utilisation
Open         | Products and services                     | 3 How inspec-
 questions   |                                           |   tion results
             |  Hydraulic valve bodies  [Sourced >]      |   are recorded
History      |  Function: flow control ...               |
             |  Customers: construction-equipment        | [Ask the
             |  OEMs ...                                 |  copilot]
```

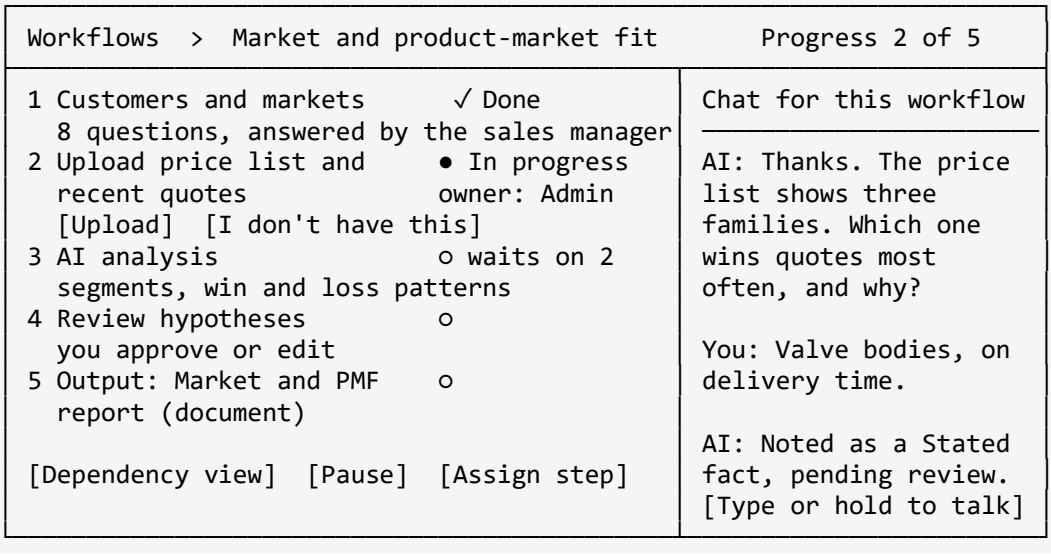


Sample generated content for the Potential section, as the AI would phrase it: "Your 5-axis capability and your OEM quality system suggest two adjacent markets: thermal-management manifolds for electric construction equipment, and valve blocks for agricultural machinery made in Kyushu. Both are inferences from your website and public OEM announcements; confirm whether you have quoted either." The "Why?" control shows the facts and sources the inference rests on.

States: a fact under review shows a dotted border; a Disputed fact shows the reporter's note; the History tab lists every change with before and after values and who made it.

5.4 Discovery workflow

A workflow page shows the steps on the left and the workflow's own chat on the right (US-D02 to US-D05). The example is the Market and product-market fit workflow at step 2.



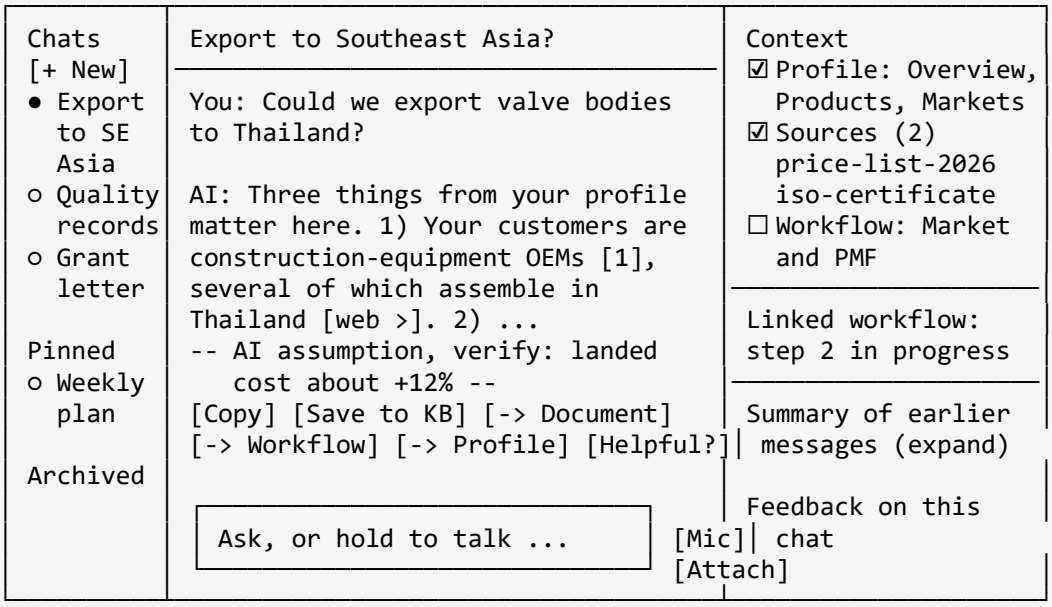
A sample question set for step 1, generated from the profile's gaps. The AI asks them a few at a time and adapts the next ones to the answers:

1. Which three customers accounted for most of your revenue last year, and roughly what share each?
2. Which product family has grown fastest in the last two years, and what do you think drove it?
3. Have you lost a quote in the last twelve months? To whom, and for what reason?
4. What is a typical lot size and lead time for valve bodies, from order to dispatch?
5. Who decides to buy at your customers: purchasing, engineering or both?
6. Which of your customers assemble outside Japan, and do they ask you to supply there?
7. Is there a product you have been asked for and declined? Why?
8. How are inspection results recorded today: paper, spreadsheet or a system?

The last question is deliberate: a market workflow also seeds the Manufacturing and AI readiness workflow, so the AI learns about the plant while asking about the market.

5.5 Chat

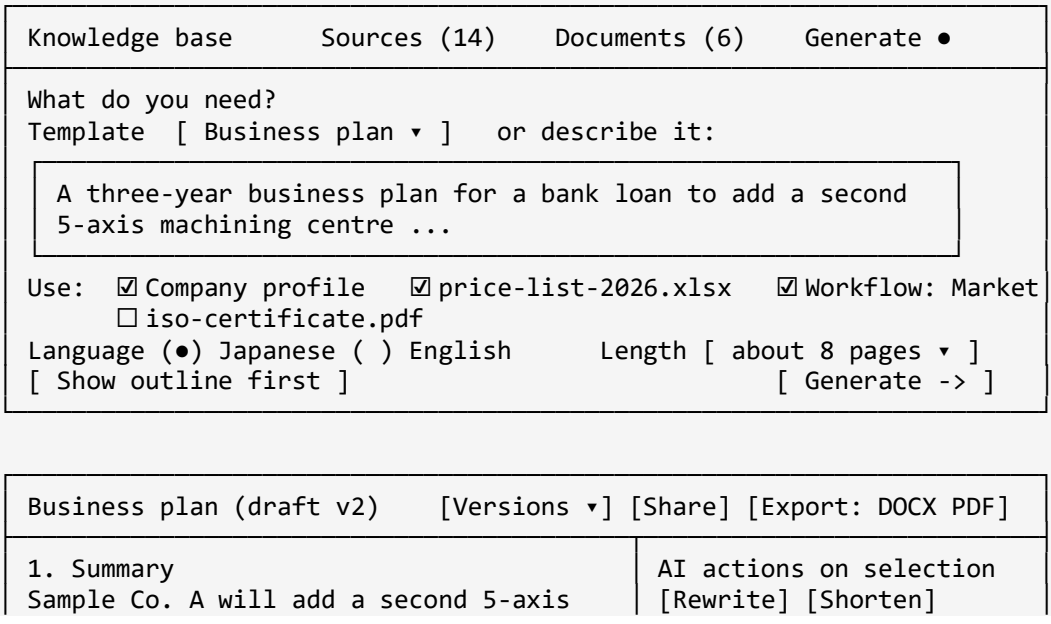
Many chats live side by side in one workspace; each shows what context the AI is using and lets the user act on any reply (US-E01 to US-E07).



States: streaming reply with a stop control; a reply given without the profile carries a notice; voice mode shows the live transcript and a "send" confirmation; the token-budget notice appears above the composer at 80%.

5.6 Knowledge base and document studio

The Generate tab turns a prompt and a template into a draft; the editor finishes it with AI actions and keeps every statement's source visible (US-F03 to US-F09).



```
machining centre in FY2027 to lift          | [Expand] [Translate JA/EN]
capacity by about 30% [AI assumption,       | [Insert profile fact]
verify] ...                                 | [Table from sources]
2. Company                                  |-----------------------------
Founded 1987 [1], 48 employees [2] ...      | Sources used
                                            | [1] website /company
3. Market                                   | [2] onboarding answer
...                                         | [3] price-list-2026
                                            |-----------------------------
                                            | Comments (2)
                                            | Feedback on this document
```

States: a source still processing is greyed in the Use list; a failed generation keeps the request so it can be retried; "Save as source" appears once a document is marked final.

5.7 AI use-case explorer

The factory map is the entry point: the seven stages in flow order with the planning, quality, maintenance, product-discovery and sales lanes around them; stages indicated by the profile are starred (US-H02 to US-H04).

```
AI use cases   Method | Factory map ● | Library | Mine (4) | Road map

 Planning and supply chain ──────────────────────────────────────────>

[Inward] > [Warehouse] > [Kitting] > [Assembly] > [Final] > [Packag-] > [Logis-]
                                        *          insp.      ing        tics

 Quality ──────────────────────>      Maintenance ──────────────────>
 R&D and product discovery <──────── Sales and product-market fit
                                                 * = indicated by profile

Assembly (machining cells, in your case)
Typical operations: operator, line leader, in-process inspector ...
Typical systems: MES, ERP production, OEE tracking, torque tools ...
Use cases here:   Recommended for you (2)    All

 Digital in-process inspection          |  Cycle-time monitoring per
 records                                |  station
 software only · H1 · L1                |  camera based · H2 · L2
 Why: you record checks on paper        |  Why: you time cycles by
 [Stated]                [+ Add]        |  stopwatch [Stated]   [+ Add]
```

A use case the company created with the wizard, as the AI drafts it (US-H05, US-H06):

```
┌─ My use case: Digital in-process inspection records ──────────────
Stage: Assembly (machining cells)      Entry point: software only
Horizon: H1 quick win                  Level needed: L1 (assisted)
Problem   Inspection results are written on paper and typed into
          a spreadsheet at shift end; first-pass yield is known two
          days late.
Change    Operators record checks on a tablet; each result links to
```

```
          the job, machine and shift; yield is visible the same
          shift.
Data      Job number, machine, operator, dimension, result, time
Systems   Today: paper and spreadsheet. Touches: ERP job cards
          (read only)
KPI       First-pass yield known the same shift (today: two days)
Creates   The inspection history that quality-drift warnings (H2)
          and closed-loop quality (H3) will need
Readiness Data ●●○  Process ●●●  People ●●○  Systems ●○○   [Check]
[Edit] [Place on road map] [Make a charter document] [Helpful?]
```

The road map board places the company's use cases on business impact against readiness to deploy, and shows the same items in horizon order underneath (US-H07):

```
          Business impact
   high
          ┌──────────────────────┬──────────────────────┐
          │ Strategic bets       │ Quick wins           │
          │ - Cycle-time         │ - Digital inspec-    │
          │   monitoring         │   tion records       │
          │ - Shortage           │ - Digital daily      │
          │   prediction         │   production plan    │
          ├──────────────────────┼──────────────────────┤
          │ Later                │ Momentum builders    │
          │ - Self-scheduling    │ - Document OCR at    │
          │   cells              │   goods receipt      │
          └──────────────────────┴──────────────────────┘
   low
           low       Readiness to deploy      high

 Programme view:  H1 quick wins --> H2 strategic bets --> H3 disruptive
                  (dependencies drawn as "needs the data created by")
```

5.8 Portfolio dashboard

The Super Admin's first screen each morning: counts, the onboarding funnel and what needs attention, all aggregated and respecting what companies have chosen to share (US-B03). Figures are sample data.

```
┌──────────────────────────────────────────────────────────────┐
│ Portfolio                                  This month ▾  [Export] │
├──────────────┬──────────────┬──────────────┬─────────────────┤
│ Companies 23 │ Active, 7 d  │ Profiles     │ Feedback positive │
│ Invited 3    │ 14 (61%)     │ approved 16  │ 84% of 212 ratings │
├──────────────┴──────────────┴──────────────┴─────────────────┤
│ Onboarding funnel                                            │
│ Invited 23 ██████████  Accepted 20 ████████  Profile 16 ███████ │
│ First workflow done 11 █████                                 │
├──────────────────────────────┬───────────────────────────────┤
│ Most-chosen use cases        │ Industries                    │
│ 1 Digital inspection records │ Metal processing 8, Food 5,   │
│ 2 Document OCR at receipt    │ Electronics 4, Plastics 3, other 3 │
│ 3 Daily plan digitised       │                               │
├──────────────────────────────┴───────────────────────────────┤
│ Needs attention: 2 dormant companies, 5 feedback items unanswered │
└──────────────────────────────────────────────────────────────┘
```

5.9 Feedback widget

The same small panel on every page (US-I02); the per-output rating controls in sections 5.3 to 5.7 feed the same inbox.

```
┌─ Feedback on this page ──────────┐
│ ( ) Something is wrong   ( ) Idea │
│ ( ) Confusing   ( ) Bug   ( ) Praise │
│ [ Tell us more ...                ] │
│ [ Attach screenshot ]               │
│ Page: /profile, JA          [Send]  │
└──────────────────────────────────┘
```

5.10 Localisation of the pages

Every page in sections 5.1 to 5.9 ships in both languages. All strings are externalised resources keyed by element, and the table lists the English strings for the shell and the chat page as the resource the developer starts from. The Japanese equivalents are produced during the build and approved by the Operator's reviewers (OQ-10); they live in the localisation file, not in this document, which is written in English for the development team.

Element	English strings	Localisation notes for the developer
Workspace navigation	Home, Company profile, Workflows, Chats, Knowledge base, Business planning, AI use cases, Feedback, Settings	Japanese labels are often wider in pixels; the rail reserves width and wraps to two lines where needed.
Top bar	Search, Notifications, Language, Ask the copilot	The language control shows each language in its own name.
Chat list	New chat, Pinned, Archived	Sort order and search must handle both scripts.
Composer	Ask, or hold to talk; Attach; Microphone	The placeholder and the send action must survive IME composition (US-K09).
Context panel	Context, Profile, Sources, Linked workflow, Summary of earlier messages	The summary is generated in the user's language.
Message actions	Copy, Save to knowledge base, Turn into document, Turn into workflow, Propose to profile, Helpful?	Icon plus label; labels collapse into a menu on phones.
Provenance labels	Stated, Sourced, Inferred, AI assumption (verify)	One agreed term each, used identically in the interface, the AI's prose and exports (Appendix F).

Element	English strings	Localisation notes for the developer
Fact controls	Confirm, Edit, Reject, Why?	Short verbs; keep button widths flexible.
Profile	Completeness, Visible to Operator, Gaps and next steps, History	"Operator" is replaced by the organisation's name in both languages.
Workflow	Progress, Done, In progress, Waits on, Assign step, Pause, Dependency view	Status words are a closed set shared with the admin console.
Document studio	Sources, Documents, Generate, Show outline first, Versions, Export	Format names (DOCX, PDF, Markdown) are not translated.
Use-case explorer	Method, Factory map, Library, Mine, Road map, Recommended for you, Add	Stage names follow the agreed terms in Appendix F.
Road map quadrants	Quick wins, Strategic bets, Momentum builders, Later	Agreed terms needed; these also appear in exported documents.
Admin console	Portfolio dashboard, Companies, Create and invite, Feedback inbox, Audit log	The admin console defaults to the Operator's language.
Feedback widget	Something is wrong, Idea, Confusing, Bug, Praise, Tell us more, Send	Categories are a closed set used in analytics; translate the label, keep the key.

A Japanese workspace sees the reply from section 5.5 in Japanese by default, generated by the same reply path from the same facts and citations. An English-language user, or a Japanese user who asks for English, sees this version:

> *Three things from your profile matter here. 1) Your customers are construction-equipment OEMs [1], several of which assemble in Thailand [web]. 2) Your ISO 9001 certification meets their local sourcing requirement [2]. 3) Landed cost is estimated at about 12% higher (AI assumption, verify).*

6. AI agent specification

The AI is one orchestrating agent that calls a small set of specialised skills and tools. Every call carries the company's verified facts first, cites what it used and leaves a trace the user can inspect. This section defines the skills, the tools, the fact ledger that is the agent's memory, how context is assembled, the guardrails and how quality is measured.

6.1 Skills

Skill	What it does	Where it appears	Main inputs	Outputs
Company analyst	Reads the website and product material, extracts facts, analyses products and potential	Onboarding, profile, re-analysis	URLs, uploads, basic facts	Proposed facts with provenance and confidence; product entries; gap list
Interviewer	Generates adaptive question sets to close gaps and interprets answers into facts	Workflows, chats	Profile gaps, question banks, prior answers	Question sets; Stated facts; follow-up questions
Workflow planner	Proposes and drafts workflows from gaps, templates or a goal; proposes what comes next on completion	Workflows, chats	Profile, templates, goal text	Workflow definitions; next-workflow proposals
Document writer	Drafts outlines and documents from a prompt, a template and retrieved sources; runs the inline edit actions	Document studio, chats	Prompt, template, selected context	Outline; draft with citations and assumption markers
Use-case matcher and assessor	Ranks library use cases against the profile, drafts charters, classifies level and horizon, runs readiness checks	Use-case explorer	Profile, library, wizard answers	Ranked recommendations with rationale; charters; readiness ratings
Planning coach (v1.1)	Pre-fills canvases, challenges assumptions, derives actions	Planning	Profile, sources, canvases	Canvas cells with provenance; critiques; tasks
Admin copilot (v1.1)	Turns natural-language admin requests into confirmed actions and answers portfolio questions	Admin console	Request text, aggregated data	Proposed action for confirmation; answers

6.2 Tools

Tool	Purpose	Side effect	Confirmation
fetch_site(url, limits)	Crawl public pages within robots.txt, depth and page limits; return text with page URLs	None	Not needed

Tool	Purpose	Side effect	Confirmation
parse_document(file)	Extract text, tables and images from uploads, with OCR when needed	None	Not needed
search_knowledge(query, scope)	Hybrid keyword and semantic retrieval over the workspace's sources, documents and chats, returning chunk citations	None	Not needed
read_profile(sections)	Return confirmed facts first, then proposed facts flagged as unconfirmed	None	Not needed
propose_fact(fact)	Add a fact to the review queue	Writes to the queue	The review queue is the confirmation
create_workflow, update_step	Create a workflow or change a step's status	Writes	Yes, in the interface
generate_document(request)	Produce an outline or a draft into the document studio	Writes a draft	Not needed; the draft is editable
match_use_cases(profile)	Score library use cases against the profile	None	Not needed
speech_to_text, text_to_speech	Voice in and out, Japanese and English	None	Not needed
web_search (restricted)	Look up public facts about markets, standards or customers named in the profile; results cited	None	Not needed; results labelled Sourced (web)
admin.create_company, admin.invite_user (v1.1)	Admin Copilot actions	Writes, sends email	Yes, explicit confirmation showing the exact values

6.3 The company fact ledger

The ledger is the agent's memory of a company. It is the only store the profile, the matcher and the writer read from, so one correction fixes everything that depends on it.

Fields: id, workspace, section, key, value, unit and period, language, provenance (Stated, Sourced, Inferred), sources (URL, document chunk, chat message or answer id), confidence from 0 to 1, status (Proposed, Confirmed, Rejected, Disputed, Superseded), created by (user or skill), timestamps, supersedes.

- Confirmed facts outrank Proposed facts; Rejected facts are never used.

- An Inferred fact is shown with its label and its "why" trail; it becomes Stated only when a user confirms it.
- Conflicts, meaning two values for one key, are surfaced in the review queue and never resolved silently.
- A number with no source is an estimate and is labelled as one.
- Deleting a source marks the facts drawn only from it as Unsourced and sends them back to review.

6.4 Context assembly for each model call

1. The policy and role prompt, versioned, including the user's language and the no-side-effect rule.
2. Workspace instructions set by the Company Admin: tone, terminology, what to avoid.
3. Confirmed profile facts for the relevant sections, then Proposed facts flagged as unconfirmed.
4. Retrieved chunks from the knowledge base with citation handles, scoped to this workspace only.
5. Workflow state (current step, prior answers) when the chat is linked to a workflow.
6. Conversation history, with older turns summarised; the summary is visible to the user.
7. The user's message and attachments.

6.5 Guardrails

- Fetched and uploaded content is data: instruction-like text in it is kept out of the instruction channel, quoted as data and flagged on the source (US-C10).
- No side effect without confirmation: tools that write outside the draft space require an interface confirmation that shows the exact values.
- No fabricated specifics: numbers, names and dates must be Stated or Sourced; otherwise the output says it is an estimate or asks.
- Scope: the AI gives business and manufacturing guidance; on legal, tax and investment matters it gives general information and recommends a professional.
- Privacy: personal data is minimised in prompts; names of people in uploads are not extracted into the profile unless a user adds them.
- Isolation: retrieval, memory and caches are partitioned per workspace; the library and the templates are the only shared content.
- Safety: content filters on input and output; a refusal explains what the AI cannot do and what it can.
- Transparency: every reply can show what it used (the context chips) and why it inferred what it inferred.

6.6 Quality and evaluation

What is measured	How	Target for release
Fact extraction precision	A golden set of 30 Japanese manufacturer websites; extracted facts checked by a reviewer	90% of extracted facts correct; every source link resolves
Question relevance	Reviewer rating of generated question sets against	4 of 5 on average

What is measured	How	Target for release
	the profile gaps	
Citation coverage in documents	Share of sourced statements that carry a resolvable citation	95%
Assumption labelling	Share of unsourced numbers that carry the estimate marker	100% in the golden set
Recommendation fit	Reviewer rating of the top five recommendations per golden profile	4 of 5 on average
User feedback	Positive rate on AI outputs from US-I01	80% after the first quarter
Injection resistance	A red-team set of crafted pages and files	No behaviour change in any case
Latency	Time to first token in chat; time to a generated profile	As in section 9

Prompts, templates and models are versioned. A change ships only after the evaluation set passes, and the previous version stays available for rollback.

6.7 Model and cost strategy

- Provider-agnostic: each skill is configured with a model tier (quality or fast) and can be switched without code changes; Japanese capability is a selection criterion.
- Embeddings are chosen for Japanese and English retrieval; a workspace's index is rebuilt when the embedding model changes.
- Website analyses and document extractions are cached per source version; repeated questions reuse retrieval results.
- Budgets: a monthly token budget per workspace (US-E09), per-call limits on context size; long outputs stream and can resume.
- Every model call is logged with references to its inputs (not their content), tokens, latency and cost, so quality and spend are traceable per feature.

7. Data model

The model has one root per company, the workspace, and keeps three kinds of things apart: what users and the AI know (facts and sources), what they do (workflows, chats, documents, use cases) and what the platform records about it (feedback, audit, consent).

Entity	Key fields	Relationships
Organisation	Name, type (Operator or Atomic Loops), default language, branding, retention policy	Has staff Users; owns Workspaces

Entity	Key fields	Relationships
Workspace	Slug, names (JA, EN), industry, size band, status, default language, token budget, sharing settings, retention	Belongs to an Organisation; owns everything below
User	Email, names, language, roles per scope, MFA, consents, notification preferences	Member of an Organisation or a Workspace; Coordinator grants
Invitation	Email, role, workspace, token, expiry, status	Created by a User
Profile section	Key, completeness, shared with Operator (yes or no)	Holds Facts
Fact	Section, key, value, unit and period, provenance, confidence, status, sources, created by, supersedes	Belongs to a Profile section; cites Sources, Messages or Answers
Product	Name, family, function, materials or process, customers, differentiators, potential	Belongs to a Workspace; has Facts
Source	Type (upload, URL, saved document, saved chat), file, extracted text, status, tags, chunks and embeddings	Belongs to a Workspace; cited by Facts, Messages and Documents
Workflow	Template and version, title, goal, status, created by (AI or user)	Has Steps, one Chat and Outputs
Step	Type, order, status, owner, due date, question set, answers, output reference	Belongs to a Workflow
Answer	Question, value, mode (text, voice, upload), answered by	Belongs to a Step; produces Facts
Chat	Title, linked workflow or document, pinned, archived, summary	Has Messages
Message	Role (user or AI), content, attachments, citations, context used, feedback	Belongs to a Chat
Document	Title, template, language, status, versions, sources used, assumption markers	Belongs to a Workspace; may become a Source
Template	Type (workflow, document, question bank), version, language fields, published	Owned by the Platform Admin or a Workspace

Entity	Key fields	Relationships
Library use case	Stage, titles and texts (JA, EN), entry point, horizon, autonomy level, data needed, systems, prerequisites, status, version	Published by the Platform Admin
Company use case	Library reference or custom, charter fields, readiness scores, quadrant, horizon, dependencies	Belongs to a Workspace
Assessment request	Status, pre-filled snapshot, contact	Belongs to a Workspace; routed to the Organisation
Feedback	Type, rating, reason, text, screenshot, page, target reference, status, reply	By a User; about a Message, Fact, Document, Use case or Page
Notification	Event, recipient, channel, read	To a User
Audit entry	Actor, action, target, workspace, time, before and after	Immutable
Consent	User, kind (terms, Operator visibility, feedback use), value, time	By a User

8. External interfaces and integrations

Everything the platform depends on outside its own code is listed here so that the provider decisions in section 12 can be made against it.

Interface	Purpose	Release	Notes
Web application	All user interaction, responsive, Japanese and English	v1	Accessibility per section 9; works in phone browsers
Large-language-model API	All generation, extraction and matching	v1	Provider-agnostic adapter; approved processing region; no training on customer data
Embedding and vector index	Retrieval over sources	v1	Partitioned per workspace
Web fetcher	Website analysis and restricted web search	v1	robots.txt and rate limits; allow-list for search
Document processing	Text, table and image extraction with Japanese OCR	v1	Runs inside the platform's own environment

Interface	Purpose	Release	Notes
Speech-to-text and text-to-speech	Voice in chats and workflows	v1.1	Japanese and English; streaming
Email	Invitations, notifications, digests	v1	Transactional provider; Japanese templates
Object storage	Uploads, exports, screenshots	v1	Encrypted; region per OQ-03
Export	DOCX, PDF, Markdown, CSV, ZIP	v1	Generated server-side
Single sign-on	Operator staff sign-in with their identity provider	v2	SAML or OIDC
Public API	Read access for the Operator's own reporting	v2	Scoped tokens
Member companies' ERP, MES and machines	Not connected	None	The platform plans use cases; execution in the plant is outside scope

9. Non-functional requirements

The platform handles company-confidential material for many competing companies at once, so isolation, privacy and provenance carry the same weight as performance.

ID	Category	Requirement
NFR-01	Performance	Chat first token within 2 seconds at p95; page loads within 2 seconds at p95 on a standard broadband connection in Japan.
NFR-02	Performance	Website analysis completes within 5 minutes for sites up to 200 pages, with live progress; a document of about 2,000 words generates within 90 seconds.
NFR-03	Capacity	v1 sized for 500 workspaces, 5,000 users and 200 concurrent users without degradation; scales horizontally beyond that.
NFR-04	Availability	99.5% monthly availability for v1, measured on the web application and chat; maintenance windows announced 48 hours ahead.
NFR-05	Backup and recovery	Daily backups; recovery point 24 hours; recovery time 8 hours; restore tested quarterly.

ID	Category	Requirement
NFR-06	Security	OWASP ASVS level 2; TLS 1.2 or higher in transit; encryption at rest; secrets in a managed vault; dependency scanning; penetration test before launch and yearly.
NFR-07	Access control	Role-based access as in section 3.3, enforced server-side; MFA for admin roles; sessions expire after 12 hours idle.
NFR-08	Tenant isolation	Every data access scoped by workspace; automated cross-tenant tests in the delivery pipeline; retrieval and caches partitioned per workspace.
NFR-09	Audit	Immutable log of admin actions and of Operator access to company data, retained 1 year; user-visible history on facts and documents.
NFR-10	Privacy	APPI compliance: purpose of use stated, consent recorded, data-subject requests handled within 30 days, personal data minimised in prompts and logs.
NFR-11	Data residency	Customer data stored in a cloud region in Japan; model processing region per OQ-04; data-processing agreements with every provider; no training on customer data.
NFR-12	Retention and deletion	Workspace retention configurable, default 24 months after deactivation; deletion requests completed within 30 days with backups expiring after; export offered before deletion.
NFR-13	Internationalisation	Japanese by default and English, switchable per user; all content, emails and AI replies in the user's language; date, number and currency formats per locale; Japanese typography and input tested.
NFR-14	Accessibility	WCAG 2.1 AA and JIS X 8341-3; keyboard operable; screen-reader labels; voice as an alternative input.
NFR-15	Usability	A Company Admin reaches an approved profile in under one hour of their own time with no training session; phone layouts for chat, workflows and feedback.
NFR-16	AI reliability	Provenance on every fact; citation and assumption markers in documents; the evaluation thresholds in section 6.6 met before each release; graceful fallback when a provider is down.

ID	Category	Requirement
NFR-17	Observability	Structured logs, traces and metrics; model-call logging without content; alerts on error rate, latency and budget.
NFR-18	Maintainability	Prompts, templates and library content versioned and editable without a deploy; feature flags per workspace; automated tests on every change.
NFR-19	Browser support	Current and previous major versions of Chrome, Edge, Safari and Firefox; Safari on iOS and Chrome on Android.
NFR-20	Cost control	Per-workspace token budgets; per-call context limits; a monthly cost report per feature for the Platform Admin.

10. Success metrics

The platform succeeds when companies keep coming back and the Operator spends minutes, not meetings, on each onboarding. The targets below are for the first full quarter after v1 and will be reset with real data.

Metric	Definition	Target after the first full quarter
Time to approved profile	From invitation sent to profile approved by the Company Admin	Median under 2 days
Profile completeness	Average completeness score across active workspaces	70% within 2 weeks of onboarding
Weekly active companies	Workspaces with at least one user session in the week	60% of active workspaces
Workflows completed	Completed workflows per active workspace per month	2
Documents generated	Documents created per active workspace per month	3
Use cases defined	Company use cases on the road map per active workspace	3
Feedback positive rate	Share of thumbs-up on rated AI outputs	80%
Feedback answered	Feedback items addressed within 5 working days	90%
Assessment requests	Requests raised through the platform	Tracked; no target in v1

Metric	Definition	Target after the first full quarter
Operator effort	Super Admin time per onboarding	Under 15 minutes

11. Release scope

Three releases, each gated on evidence from real companies rather than on a date: v1 is the smallest product the Operator can onboard companies with, v1.1 adds voice, planning and the lite assessment, and v2 opens the platform to programmes, identity providers and other organisations.

Release	Scope	Exit gate
v1	Epic A; Epic B (US-B01 to B04, B06, B07); Epic C; Epic D (US-D01 to D05, D08); Epic E except voice; Epic F (US-F01 to F03, F05 to F07, F09); Epic H (US-H01 to H05, H07, H10); Epic I (US-I01 to I06); Epic J (US-J01, J03, J04, J06); Japanese and English; evaluation thresholds met	Five pilot companies onboarded by the Operator; positive feedback rate of 70% in the pilot; security review passed
v1.1	Voice (US-E04); planning canvases (Epic G); readiness check (US-H06); lite assessment and assessment request (US-H08, H09); Admin Copilot (US-B08); announcements (US-B09); workflow template editor (US-B05); the remaining Should and Could stories	Pilot companies using voice and planning; the lite assessment produces a preliminary summary that Atomic Loops has reviewed
v2	Programmes and connections (US-B10); single sign-on; public API; whole-document translation (US-F10); support for a second organisation (another prefecture or agency)	The Operator confirms the programme-matching rules; a second organisation is signed

What is deliberately not in any release: connecting to member companies' ERP, MES or machines, and running AI use cases in the plant. Those remain the work of an Atomic Assessment and its follow-on projects.

12. Open questions and decisions needed

Fifteen decisions shape the build; the first four block the architecture and should be settled with the Operator before design starts.

- ☐OQ-01 Product name, Japanese name and branding: the Operator's brand with "built with Atomic Loops", or a joint brand?
- ☐OQ-02 Who is the legal operator of the service and the data controller under APPI: the Operator, with Atomic Loops as processor?
- ☐OQ-03 Hosting provider and region for data at rest (a region in Japan is assumed).

- ☐OQ-04 Approved large-language-model provider and processing region, with confirmation that its terms exclude training on customer data.
- ☐OQ-05 Speech provider for Japanese voice (v1.1).
- ☐OQ-06 Email sender identity: the Operator's domain or the platform's.
- ☐OQ-07 Default sharing with the Operator: overview and products summary only (proposed), or more?
- ☐OQ-08 Does the Coordinator role match how the Operator works today, and should it be in v1?
- ☐OQ-09 Expected number of companies in year one and the token budget per workspace.
- ☐OQ-10 Japanese review of the library and explainer content: who at the Operator reviews it, and how often?
- ☐OQ-11 Is the lite assessment positioned as preparation for an Atomic Assessment, or as a stand-alone deliverable the Operator presents?
- ☐OQ-12 Support model: who answers company questions in Japanese, and in which hours?
- ☐OQ-13 Sign-in: email and password with MFA (proposed), or the Operator's identity provider from v1?
- ☐OQ-14 Terms of use and privacy notice: drafted by whose counsel?
- ☐OQ-15 Next step: a clickable prototype of the onboarding wizard, the profile and the use-case explorer (sections 5.2, 5.3 and 5.7) before build, to test with two or three member companies?

13. Appendices

Appendix A: use-case library seed

The starting content of the library, condensed from the Atomic Loops Factory Assessment Framework and the Atomic Assessment method. Each row becomes a card (US-H03). The library spans every stage and grows after launch; value ranges are added only where Atomic Loops has measured them and never name a client.

Stage or lane	Use case	Entry point	Horizon	Level	Data it needs
Step 0, all stages	Factory intelligence dashboard: one source of truth with a view per role (leadership, production, quality, maintenance, R&D, supply chain, sales)	Software only	H1	L1	Records already in ERP and spreadsheets
Inward	Document reading at the gate: invoice, delivery note and test certificates read and matched to the purchase order	Software only	H1	L1	Purchase orders, scanned documents
Inward	Material test certificate check	Software	H1	L1	Certificates,

Stage or lane	Use case	Entry point	Horizon	Level	Data it needs
	against the specification library	only			specification thresholds
Inward	Gate-to-receipt cycle time tracking by QR scan	Software only	H1	L1	Arrival and receipt timestamps
Inward	Supplier risk scoring from delivery and quality history	Software only	H2	L2	Delivery performance, rejection history
Warehouse	Part identification from a photo for unlabelled material	Camera based	H2	L2	Master product images
Warehouse	Put-away bin recommendation	Software only	H2	L2	Item dimensions, usage frequency, bin master
Warehouse	FIFO exception alerts and inventory anomaly detection	Software only	H1	L1	Lot dates, stock movements
Warehouse	Demand-driven replenishment ahead of a line shortage	Software only	H2	L3	Production schedule, consumption rates
Kitting	Kit completeness check before dispatch to the line	Camera based	H2	L2	Bill of kit, kit images
Kitting	Shortage prediction hours ahead of a kit stop	Software only	H2	L2	Stock, open orders, consumption history
Kitting	Lot-to-kit traceability	Software only	H1	L1	Lot numbers, kit numbers, serials
Assembly or machining	Digital in-process inspection records	Software only	H1	L1	Job, machine, operator, result
Assembly or machining	Cycle-time monitoring per station against takt	Camera based or machine IoT	H2	L2	Cycle start and end per unit
Assembly or machining	Pick-and-place and assembly verification	Camera based	H2	L2	Station images, work instructions

Stage or lane	Use case	Entry point	Horizon	Level	Data it needs
Assembly or machining	OEE and bottleneck finder	Machine IoT	H2	L2	Machine states, cycle logs, downtime reasons
Assembly or machining	Andon root-cause suggestions	Software only	H2	L2	Andon history by station, shift and model
Assembly or machining	Predictive maintenance on critical machines	Machine IoT	H3	L3	Vibration, temperature and current history
Assembly or machining	Visual defect inspection against golden samples	Camera based	H2	L2	Labelled defect images
Final inspection	Mobile inspection app with enforced photo evidence	Software only	H1	L1	Checklists, serials
Final inspection	Automatic inspection report and certificate generation	Software only	H1	L1	Checklist results, test data
Final inspection	Non-conformance pattern analysis and corrective-action suggestions	Software only	H2	L2	NCR history, CAPA records
Final inspection	Dispatch hold linked to the inspection outcome	Software only	H1	L1	Inspection result, delivery order
Packaging	Label verification against master data	Software only	H1	L1	Label fields, ERP master
Packaging	Wrong-pack detection by weight or image	Camera based	H2	L2	Pack specifications, weights
Packaging	Serialised traceability from raw lot to customer order	Software only	H1	L1	Lot, serial, carton, order
Logistics	Load plan optimisation per truck	Software only	H2	L2	Carton dimensions, orders, departure times
Logistics	On-time delivery prediction and proactive customer alerts	Software only	H2	L2	GPS, route history, delivery windows

Stage or lane	Use case	Entry point	Horizon	Level	Data it needs
Logistics	Proof-of-delivery reading and validation	Software only	H1	L1	POD photos, delivery notes
Planning and supply chain	Daily production plan digitised with change history	Software only	H1	L1	Plan spreadsheets
Planning and supply chain	Finite-capacity scheduling on top of the ERP plan	Software only	H2	L3	Routings, capacities, orders
Planning and supply chain	Demand forecasting at warehouse and SKU level	Software only	H2	L2	Sales history, distributor data
Planning and supply chain	Self-scheduling lines with planner approval of exceptions	Software only	H3	L4	All of the above, matured
Quality	Automated quality Pareto and first-pass-yield dashboard	Software only	H1	L1	Rejection records
Quality	Quality-drift warnings and root-cause suggestions	Software only	H2	L2	Inspection history linked to machine, batch and shift
Quality	Closed-loop quality with parameter correction	Machine IoT	H3	L4	Parameters, outcomes, drift history
Maintenance	Breakdown notification linked to production loss	Software only	H1	L1	Breakdown tickets, output records
Maintenance	Self-tuning machine parameters	Machine IoT	H3	L4	Parameter and outcome history
R&D and product discovery	Market and complaint signals as new-product triggers	Software only	H2	L2	Sales, complaints, market sources
R&D and product discovery	Bill-of-material and cell-change impact simulation	Software only	H2	L2	BOM, routings, cell data
R&D and product discovery	Product-market fit hypothesis tracking with evidence	Software only	H1	L1	Profile, sources, workflow answers

Stage or lane	Use case	Entry point	Horizon	Level	Data it needs
Sales and product-market fit	Must-sell recommendations per customer or outlet	Software only	H2	L2	Sales by customer and product
Sales and product-market fit	Sales pitch and role-play training built from product and application knowledge	Software only	H1	L1	Product documents
Sales and product-market fit	Competitor and review intelligence	Software only	H2	L2	Public sources

Appendix B: workflow templates

Each template is a numbered sequence of steps of the five types in US-D02. Steps marked (AI) run without the user; the others need the company. Templates are versioned by the Platform Admin (US-B05).

B1, Company discovery, proposed to every new workspace:

1. Confirm the generated overview (review).
2. Question set: people and roles, plants and shifts, certifications, main customer types (about 10 questions).
3. Upload, optional: product catalogue, organisation chart, a recent company introduction.
4. (AI) Analysis: gaps, inferred market position, proposed products list.
5. Review the proposed facts.
6. Output: profile version 1 approved; the AI proposes the next workflows.

B2, Product deep-dive, one per main product family:

1. Question set: function, variants, materials and process, volumes and lot sizes, pricing position, who buys and why.
2. Upload, optional: drawings, specification sheet, photos.
3. (AI) Analysis: differentiators, adjacent applications, risks, what the product needs from the plant.
4. Review.
5. Output: a product one-pager; the product entry in the profile is updated.

B3, Market and product-market fit, shown in section 5.4:

1. Question set: customers and markets.
2. Upload: price list and recent quotes.
3. (AI) Analysis: segments, win and loss patterns, demand signals.
4. Review the hypotheses.
5. Output: market and product-market fit report; hypotheses go to the planning board (v1.1).

B4, Manufacturing and AI readiness:

1. Question set: lines, machines, shifts, systems in use.
2. Stage-by-stage capture: which stages exist and how the result is recorded at each (paper, spreadsheet, system, sensor).
3. Pain-point interview in the workflow's chat.
4. (AI) Analysis: recommended use cases, readiness gaps, the data each quick win would create.
5. Review.
6. Output: use-case candidates placed on the road map.

B5, Business plan:

1. Goal and audience: bank, customer or a prefecture programme.
2. Question set: targets, planned investments, headcount, financing.
3. Context: profile, canvases and selected sources.
4. Outline approval.
5. (AI) Draft, then edit.
6. Output: business plan document.

B6, Export readiness:

1. Target markets.
2. Question set: certifications, logistics, pricing, language, partners.
3. (AI) Analysis: gaps and the partners needed.
4. Output: export readiness memo.

B7, AI use-case charter:

1. Wizard (US-H05).
2. Readiness check (US-H06, v1.1).
3. Output: charter document and placement on the road map.

B8, Lite assessment (v1.1):

1. KPI alignment questions.
2. Pain-point interview.
3. Capture per line or machine: the data lane, the human lane and the material lane.
4. Document checklist.
5. (AI) Analysis.
6. Output: preliminary readiness summary with recommended quick wins.
7. Optional: request an Atomic Assessment.

Appendix C: sample question bank

Three samples of the banks the Interviewer skill draws from; the AI picks and rewords questions according to the gaps and the earlier answers.

B1, Company discovery:

- How many people work at the company, and how many of them on the shop floor?

- Which plants or sites do you run, and what does each make?
- Which certifications do you hold, and since when?
- Who are your typical customers: OEMs, tier suppliers, distributors or end users?
- What has changed most in your business in the last three years?

B4, Manufacturing and AI readiness:

- Which of these stages exist in your plant: goods receipt, warehouse, kitting, assembly or machining, final inspection, packaging, dispatch?
- At each stage, how is the result recorded: on paper, in a spreadsheet, in a system or by a sensor?
- Which machine produces data that nobody looks at?
- Which decision in the plant takes longest to reach, and who makes it?
- What is counted by hand today that a system or a camera could count?
- When a defect is found, how long is it until the cause is known?

B5, Business plan:

- What is the plan for, and who will read it?
- What investment are you planning, and when?
- What do you expect it to change: capacity, quality, lead time or new products?
- How will it be financed, and over how many years?
- Which assumptions worry you most?

Appendix D: document templates in v1

- Company introduction
- Business plan
- Product one-pager
- Market analysis
- Product-market fit report
- Export readiness memo
- Competitor comparison
- Programme or subsidy application narrative (the company's description of itself and its plan, in the form the Operator's programmes ask for)
- AI use-case charter
- Factory AI road map
- Customer proposal
- Meeting preparation brief, for the company or a Coordinator before a meeting

Appendix E: traceability from the brief to the stories

Brief item (section 1.2)	Delivered by
1 Super-admin access and logins for new companies	US-A01 to A03, A10; US-B01, B02, B08
2 Website and product details analysed by the AI	US-C01 to C03, C07

Brief item (section 1.2)	Delivered by
3 Follow-up questions and follow-up workflows	US-D01 to D09; US-E07
4 The company's AI-generated page	US-C04 to C06, C08, C09
5 Many chats, seeing the workflow, talking to the AI	US-E01 to E10; US-D05
6 Knowledge base, prompts to documents, a copilot for business planning	US-F01 to F10; US-G01 to G05
7 Feedback everywhere	US-I01 to I07
8 Use cases from the audit workflow, product discovery and product-market fit, creating your own	US-H01 to H10
9 Japanese first, English equal, across dashboards, chats and generative AI	US-K01 to K11; US-A09; NFR-13

Appendix F: localisation glossary of stages and terms

The stage names and method terms that need one agreed Japanese equivalent each before launch. The equivalents are agreed with the Operator's reviewers (OQ-10) and stored in the localisation resource and the workspace glossary; this document stays in English for the development team.

Term	Where it is used	Note for the Japanese equivalent
Goods receipt (inward)	Factory map, library, lite assessment	Use the term the member companies' own documents use; "inward" is Atomic Loops' label.
Incoming inspection	Library	Standard quality-management term.
Warehouse	Factory map, library	Plain term.
Kitting	Factory map, library	Often a loan word in Japanese plants; confirm with the Operator.
Assembly; Machining	Factory map	Machining is the variant of the assembly stage shown to job shops.
In-process inspection	Library	Standard term.
Final inspection (pre-delivery inspection)	Factory map, library	The deck's abbreviation PDI is spelled out everywhere.
Packaging	Factory map	Plain term.
Dispatch and logistics	Factory map	Plain term.
Production planning and	Lane on the factory	Plain terms.

Term	Where it is used	Note for the Japanese equivalent
supply chain	map	
Quality; Maintenance	Lanes on the factory map	Plain terms.
Predictive maintenance	Library	Established term.
R&D and product discovery	Lane on the factory map	"Product discovery" needs an agreed rendering; product planning is the usual neighbour.
Product-market fit	Lane, planning canvases	Usually a loan word; confirm.
Sales	Lane on the factory map	Plain term.
Gemba walk	Method explainer	Japanese in origin; use the plant's own everyday term.
Quick win; Strategic bet; Disruptive solution	Method explainer, road map	Three agreed terms, used identically in the interface and in exports.
Autonomy level L0 to L5: Manual, Assisted, Partial, Conditional, High, Full	Method explainer, use-case cards	Keep the L numbers; translate the six names.
Readiness; Entry point (software only, camera based, machine IoT)	Use-case cards, road map	Agreed terms.
Company profile; Workflow; Knowledge base; Road map; Feedback	Navigation	Agreed terms; some may stay loan words.
Stated; Sourced; Inferred	Provenance labels everywhere	Three short agreed terms; the AI uses them in its prose too.